

Introduction to LaTeX

You are encouraged to work with your neighbors on all in class activities.

Learning Objectives: *Learn the basics of how LaTeX works, including how to set up a LaTeX document, how to typeset text, typeset equations, and handle compile errors. Learn how to use core concepts, such as commands, environments, and packages. Be able to edit a Latex template to make your own paper.*

LaTeX is a powerful and flexible writing environment that makes beautiful documents, especially when using mathematics. It was created by scientists, for scientists, and has a large and active community. LaTeX is powerful — you can extend it by using special packages for papers, presentations, spreadsheets, and more.

1. **Getting started.** Go to the Overleaf website (<https://www.overleaf.com>) and click the login button. Next, scroll down and click *Log in through your institution*. Complete the process to login through your UT email or EID, confirm through duo mobile if required and agree to FERPA rules. Note: you can use Overleaf for free without UT authentication but using the UT access unlocks some additional features not available in the free version.

2. **Create a new LaTeX project.** We will start by making a new “project” from scratch:

- In the main page click the *New Project* button, then click *Blank Project* from the drop down menu. A box will open where you should enter the name of your new project, then click *Create*.

You should now see some basic LaTeX commands:

```
1 \documentclass{article}
2 \usepackage{graphicx} % Required for inserting images
3
4 \title{AST376R Project}
5 \author{Stella Offner}
6 \date{February 2024}
7
8 \begin{document}
9
10 \maketitle
11
12 \section{Introduction}
13
14 \end{document}
15
```

- Comment out different lines using the % symbol and then run the code by clicking the *Recompile* button to see what they do. Write a sentence after the `\section{Introduction}` command and *Recompile*.

You can see that LaTeX automatically indents the first line of the paragraph, taking care of the formatting for you.

3. Document Components.

There are several different parts of the document:

Document Class: The first line of code, `\documentclass{article}`, declares the document type known as its class, which controls the overall appearance of the document. Different types of documents require different classes, i.e., a CV/resume will require a different class than a scientific paper which might use the standard LaTeX article class. Other types of documents you may be working on may require different classes such as book or report.

- To get some idea of the many LaTeX class types available, visit the relevant page on CTAN (Comprehensive TeX Archive Network): <https://ctan.org/topic/class>

Packages: Like Python, LaTeX contains pre-created packages that extend the basic functionality of the environment. Here, `\usepackage{graphicx}` includes a set of commands that are needed for inserting plots into the document.

- To explore different packages, visit the CTAN list of package: <https://ctan.org/pkg>. Browse the (very long) list of options. What interesting ones do you see?

Body: The paper content, known as the body of the document, is written between the `\begin{document}` and `\end{document}` tags.

4. **Preamble.** The menu on the far left shows Overleaf storing a LaTeX document as a file called *main.tex*, where the .tex file extension is, by convention, used when naming files containing your document's LaTeX code.

The previous example showed how document content was entered after the `\begin{document}` command; however, everything in your .tex file appearing before that point is called the *preamble*, which acts as the document's *setup* section. Within the preamble you define the document class (type) together with specifics such as

- Languages to be used when writing the document
- Packages you would like to use (like *graphicx*)
- Apply other types of configurations
- Define useful shorthand commands (like mathematical variables)
- and more ...

A minimal document preamble might look like this:

```
\documentclass[12pt, letterpaper]{article}  
\usepackage{graphicx}
```

- Add these options to your *documentclass* line. Experiment with changing the 12pt number. What does it do?
- Add an new *twocolumn* argument to the document class arguments. What does it do?

5. Latex RNAAS Template. Let's experiment with a more complex LaTeX example template.

- Go back to your “home” page (icon in the top left). Click *New Project* button, then *Upload Project*. Drag and drop the *rnaas.zip* folder (download from Canvas, in Module 4).
- Read through the document. What is the purpose of RNAASs? What is required for a RNAAS? What distinguishes it from a regular AAS paper?
- Keep this template, you will use it for Homework 5.